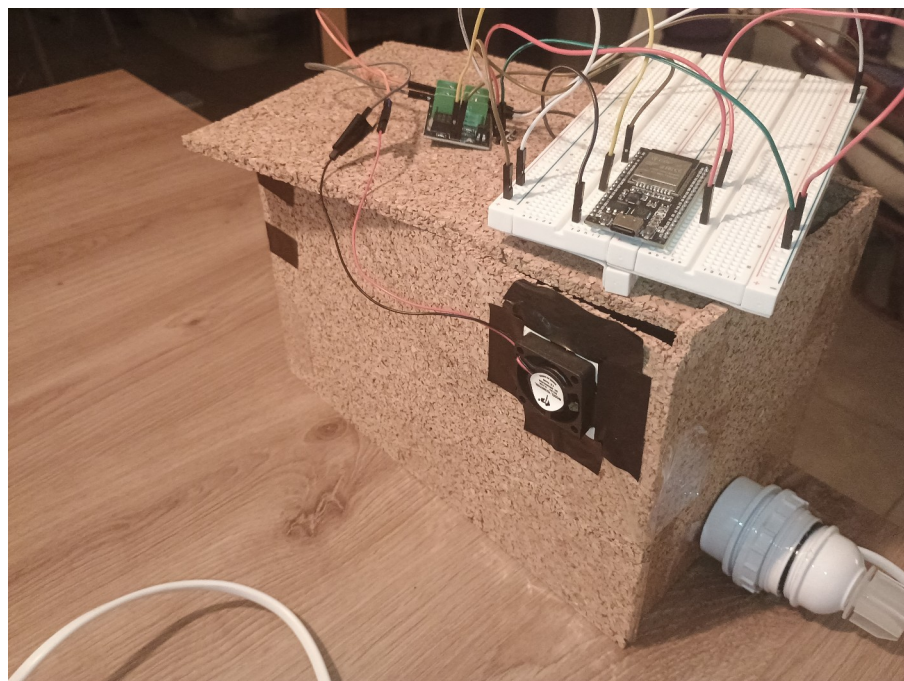
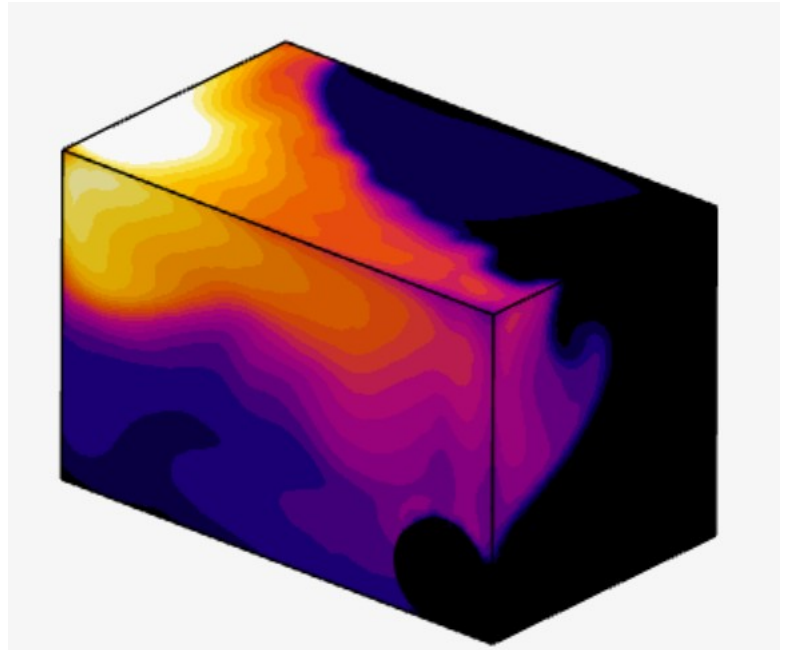


*Design, Implementation, and Validation of a Safety-Oriented
Closed-Loop Thermal Management System based on ESP
Micro-controller*



1. Introduction

1.1 Scope

The scope of this project is the design, implementation, and validation of a small thermal enclosure containing a constant heat source. The internal temperature is regulated by an active control system based on an ESP32 microcontroller.

The control system implements a discrete **PI controller with rate-of-change setpoint Feed-Forward (FF)**. The controller is designed to regulate the chamber temperature both around constant setpoints and along dynamic temperature profiles.

The project follows a Systems Engineering workflow in which system requirements, design elements, safety functions and verification methods are systematically defined and traced throughout the development lifecycle. The overall development process follows the principles of the **V-Model**, with requirements and safety objectives defined during the design phase and subsequently verified and validated through analysis, simulation, and experimental testing.

A functional-safety-oriented approach is also adopted. The methodology is derived from the principles of **IEC 61508**, the general standard for the functional safety of electrical, electronic, and programmable electronic safety-related systems. Standards such as ISO 26262 for automotive systems and IEC 61513 for nuclear applications are based on similar principles.

Since the present system is a laboratory prototype rather than a safety-certified industrial product, a tailored methodology inspired by these standards is applied. A top-down **Hazard Analysis and Risk Assessment (HARA)** is combined with a bottom-up **Failure Mode and Effects Analysis (FMEA)**. This approach is used to identify the principal hazard, derive a Safety Goal and Safety Requirements, and verify that the architecture provides sufficient protection against the identified failure modes without introducing the full administrative framework required by a certification process.

The project does not claim compliance or certification according to IEC 61508. Instead, selected concepts from functional-safety engineering—including hazard analysis, safety goals, safety requirements, fault analysis, independence, safe-state definition, and verification—are adapted to the scope of the laboratory prototype

1.2 Limitations

The physical enclosure is fabricated using basic manual tools. Consequently, the mechanical fabrication tolerances are moderate and the enclosure is not intended to represent a production-quality thermal chamber.

Nevertheless, the experimental thermal behaviour shows a good correlation with the preliminary one-dimensional thermal models. A three-dimensional Conjugate Heat Transfer (CHT) CFD model is additionally used to investigate the spatial temperature distribution and airflow behaviour and to optimize the placement of the fan and temperature sensor.

The application of functional-safety principles is intentionally tailored to the prototype nature, scale, and intended operating environment of the system.

2. System Requirements

2.1 Functional Requirements

REQ-001 — Static temperature control

The system shall be capable of maintaining the enclosure temperature within $\pm 1^{\circ}\text{C}$ range of a constant temperature setpoint.

REQ-002 — Dynamic temperature control

The system shall be capable of tracking a varying temperature setpoint within $\pm 1^{\circ}\text{C}$ error range.

REQ-003 — Real-time graphical representation

The system shall provide the user with a graphical representation of the measured temperature, temperature setpoint, and actuator utilisation in real time.

REQ-004 — Tabular data

The system shall provide measured temperature and actuator utilisation in tabular form for data logging and subsequent analysis.

2.2 Assembly Requirement

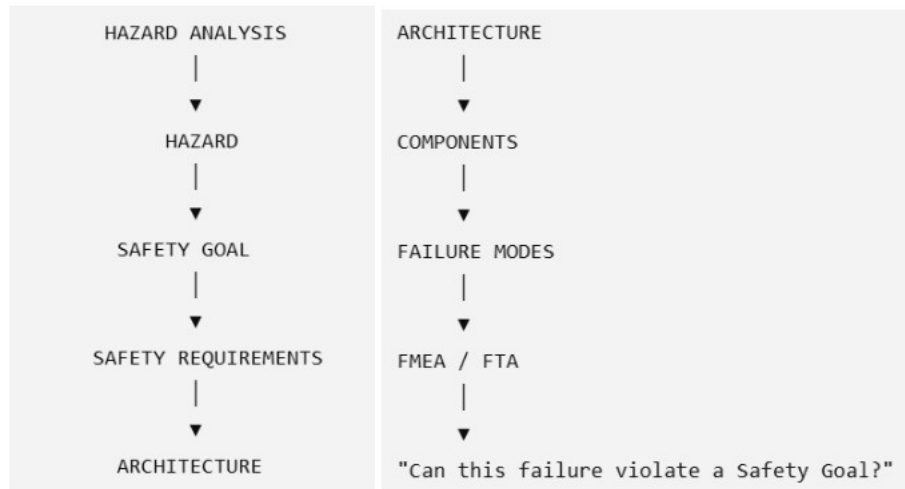
REQ-005 — Modular assembly

The system shall be assembled using, as far as reasonably possible, modular and plug-and-play components in order to minimize permanent connections and avoid unnecessary soldering.

2.3 Safety Requirements

The safety requirements are derived using a methodology inspired by the top-down and bottom-up approaches commonly used in functional-safety engineering.

The top-down analysis begins with the identification of the principal hazard and the corresponding Safety Goal. The Safety Goal is subsequently decomposed into technical Safety Requirements. FMEA is then used to verify that the identified failure modes are adequately addressed by the architecture.



Hazard Analysis

HAZARD-01 — Uncontrolled overheating

Uncontrolled overheating of the enclosure could result in damage to components, fire, or injury.

Safety Goal

SG-01

The system shall prevent the temperature of the electronic components and sensor region of the enclosure from exceeding the defined maximum allowable temperature TMAX.

The value of TMAX is derived from the operating limits of the components installed inside the enclosure and from the intended operating environment.

Safety Requirements

SR-01 — Periodic temperature monitoring

The control system shall acquire and validate the enclosure temperature at intervals not exceeding the maximum monitoring interval specified for the temperature sensor during active operation.

If a valid temperature measurement is not received within the defined monitoring timeout, the system shall identify the condition as a temperature measurement fault and initiate the response specified by SR-02 and SR-04.

SR-02 — Temperature measurement failure detection

The control system shall detect loss or invalidity of the temperature measurement.

A temperature measurement shall be considered invalid when sensor communication fails, the received data fail the implemented data-validation mechanism, or the measured value is outside the specified valid operating range of the sensor.

Detection of a temperature measurement failure shall cause the system to transition to STATE_FAULT in accordance with SR-04.

SR-03 — Cooling actuator failure tolerance

Loss of the cooling actuator shall not result in violation of SG-01 when the system operates within the environmental operating envelope.

The safety claim shall not depend on successful detection of the cooling-actuator failure if the independent passive thermal protection specified by SR-05 is demonstrated to maintain $T_{enclosure} < T_{MAX}$ under the corresponding worst-case thermal condition.

SR-04 — Safe-state transition

When a critical temperature measurement fault or critical control-system fault is detected, the system shall transition to a predefined safe state.

In STATE_FAULT, the system shall command the cooling actuator to its defined safe-state output of 100% duty cycle, where the actuator and power architecture permit this operation.

SR-05 — Independent passive thermal protection

An independent passive thermal protection mechanism shall limit the temperature below T_{MAX} in the event of loss of the active temperature-control function.

The passive protection shall remain effective in the event of failure of the temperature sensor, control software, microcontroller, cooling actuator, or associated active-control functions.

The passive thermal protection shall be demonstrated for the worst-case heat input and for all ambient temperatures within the operating envelope

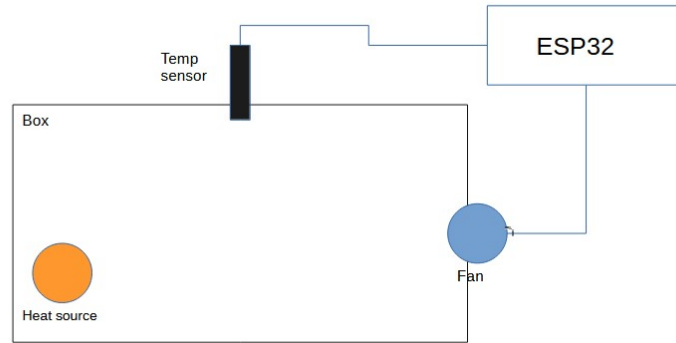
SR-06 — Fault detection and response time

The system shall detect a loss or invalidity of the temperature measurement within a defined maximum fault-detection time.

Following detection of a critical temperature measurement fault, the system shall enter STATE_FAULT and issue the defined safe-state actuator command within the specified maximum fault-response time. The maximum detection and response times shall be compatible with the thermal dynamics of the enclosure and shall be verified experimentally.

2.4 System Architecture

The proposed architecture consists of commercially available, low-cost components selected to satisfy the modularity requirement REQ-005.



The principal elements are:

- ESP32 microcontroller
- DHT22/AM2302 temperature sensor
- 5 V cooling fan
- MOSFET fan driver
- Constant 15 W heat source (bulb)
- Cork thermal enclosure
- Breadboard and Dupont connections
- Wi-Fi communication link
- PC-based graphical interface and data logger

The architecture combines active temperature regulation with passive thermal protection. The active control system is responsible for maintaining the desired temperature profile during normal operation, while the passive thermal characteristics of the enclosure provide an independent limitation of the maximum temperature (see 2.6 Passive Safety) in the event of active-system failures.

2.5 Failure Mode and Effects Analysis

A basic FMEA was performed to identify the principal failure modes affecting the thermal-control system.

Failure Mode	Operational Impact	Safety Impact	Safety Mechanism
FM-01:	Sensor disconnected or	No violation of SG-01 because	SM-01: Software fault

Failure Mode	Operational Impact	Safety Impact	Safety Mechanism
Temperature sensor failure	invalid data results in loss of feedback and prevents normal control-loop computation.	the passive thermal boundary limits the maximum temperature.(See 2.6 Passive Safety chapter)	handling commands the fan to 100% and enters the fault state when sensor data are invalid.
FM-02: Fan failure	Loss of airflow increases the operating temperature of the enclosure.	No violation of SG-01 within the specified operating envelope because passive heat dissipation limits the maximum temperature.(See 2.6 Passive Safety)	SM-02: Passive thermal limit provides independent protection.
FM-03: Control-unit failure	A software freeze or processor failure may leave the actuator command unchanged.	No violation of SG-01 within the specified operating envelope because passive heat dissipation limits the maximum temperature. (See 2.6 Passive Safety)	SM-03: Hardware Watchdog Timer resets the controller following software failure.

The distinction between **operational impact** and **safety impact** is important. A failure can significantly degrade the availability or performance of the temperature-control function while remaining non-critical from a thermal-safety perspective because the passive thermal design independently limits the maximum temperature.

2.6 Passive Safety

The passive safety argument is based on the maximum allowable temperature of the components and the thermal behaviour of the enclosure.

The relevant component temperature ratings are:

Component	Operating Temperature Range
Fan	-20°C to 80°C
DHT22 temperature sensor	-40°C to 80°C
Cork enclosure material	-20°C to 120°C

The most restrictive relevant electronic component limit is therefore:

TMAX=80°C

The thermal model predicts that, under the worst active-component failure considered, i.e. operation without forced airflow, the enclosure temperature remains bounded by approximately

$$T_{max} \approx T_{env} + 19^{\circ}\text{C}$$

within the specified operating environment.

Therefore, for the nominal indoor operating envelope, the passive thermal behaviour provides an independent protection mechanism against the identified overheating hazard.

The CFD model also shows local temperature peaks near the heat source. These peaks are associated with the idealized representation of the heat source in the simulation and are not representative of the temperature of the electronic components. In the physical implementation, the heat source is not in direct contact with the sensor, fan, or electronic components.

In particular with reference to SG-01 we define

Protected thermal zone:

- ESP32
- DHT22
- MOSFET
- fan

and Non-protected heat-source region:

- bulb
- immediate bulb surroundings
-

Consequently, the passive-safety argument must be understood as applying to the **temperature boundary relevant to the components and enclosure**, rather than to the idealized local CFD temperature immediately adjacent to the heat-source surface.

2.7 Single Point of Failure and Common Cause Failure Analysis

Single Point of Failure

No single identified failure of the temperature sensor, fan, or control unit is expected to result in a violation of SG-01 within the defined operating envelope.

This is primarily achieved through the passive thermal characteristics of the enclosure. The loss of active cooling or active control therefore results in degraded thermal performance but does not, within the specified boundary conditions, cause the temperature to exceed the defined safety limit.

Common Cause Failures

The principal common-cause failure conditions identified are:

1. **Power supply loss**

Loss of the power supply simultaneously disables the ESP32 and the fan. The passive thermal characteristics of the enclosure remain available and therefore continue to provide the independent thermal boundary.

2. Excessive ambient temperature

The passive thermal safety argument depends on the ambient temperature remaining within the specified operating envelope. An ambient temperature above 61°C would reduce or eliminate the passive safety margin.

The enclosure is intended for indoor laboratory operation, and such an ambient temperature is considered outside the specified operational envelope.

3. Design and Implementation

3.1 Thermal Box Design

The first step in the thermal design is to determine the steady-state temperature under the two limiting airflow conditions:

- Fan OFF
- Fan at 100% duty cycle

These two cases define the thermal operating envelope within which the active controller must operate.

3.1.1 Fan-Off Case

For the fan-off case, the enclosure is initially modelled using a one-dimensional steady-state wall heat-transfer model.

The principal parameters are:

- Heaterpower:
Q=15 W
- Cork wall thermal resistance:
R_{wall}=0.175 m²K/W
- External convection coefficient:
h≈5 W/(m²K)
- Total external surface area:
A=0.184 m²

$$R_{\text{cond}} = \frac{L}{kA}$$

$$R_{\text{conv}} = \frac{1}{hA}$$

The steady-state temperature difference is obtained from the thermal resistance between the heat source and the external environment.

$$T_{\text{steady}} = T_{\text{amb}} + Q R_{\text{tot}}$$

$$R_{\text{tot}} = R_{\text{wall,int}} + R_{\text{wall}} + R_{\text{conv,ext}}.$$

$$T_{\text{steady}} = T_{\text{amb}} + 19$$

3.1.2 Fan at 100% Case

For the forced-convection case, the steady-state temperature can be estimated using an energy balance over the enclosure control volume.

The heat introduced by the heater is balanced by the heat carried away by the airflow and the conduction:

$$P_{\text{heater}} = \left(\dot{m}_{\text{air}} \cdot c_p + \frac{k \cdot A}{L} \right) (T_{\text{regime}} - T_{\text{amb}})$$

where:

- $P_{\text{heater}} = 15 \text{ W}$
- $c_p = 1005 \text{ J/(kg K)}$
- Maximum fan volume flow:
 $\dot{V} = 5 \text{ m}^3/\text{h} = 0.0014 \text{ m}^3/\text{s}$
- Air density: $\rho = 1.2 \text{ kg/m}^3$

The corresponding mass flow rate is:

$$\dot{m} = \rho \dot{V}$$

$$\dot{m} = 1.2(0.00138889)$$

$$\dot{m} \approx 0.0017 \text{ kg/s}$$

And T at steady state is equal to:

$$T_{\text{regime}} = T_{\text{amb}} + \frac{P_{\text{heater}} \cdot L}{\dot{m}_{\text{air}} \cdot c_p \cdot L + k \cdot A}$$

$$T_{\text{steady state}} = T_{\text{amb}} + 8.00^\circ\text{C}$$

Finally the operating range of temperatures for the chamber is

$$T_{\text{amb}} + 8.00^\circ\text{C} \leq T_{\text{chamber}} \leq T_{\text{amb}} + 19.00^\circ\text{C}$$

3.2 CFD Simulation

A steady-state three-dimensional Conjugate Heat Transfer (CHT) analysis was performed using the CAD model.

The objectives of the simulation were:

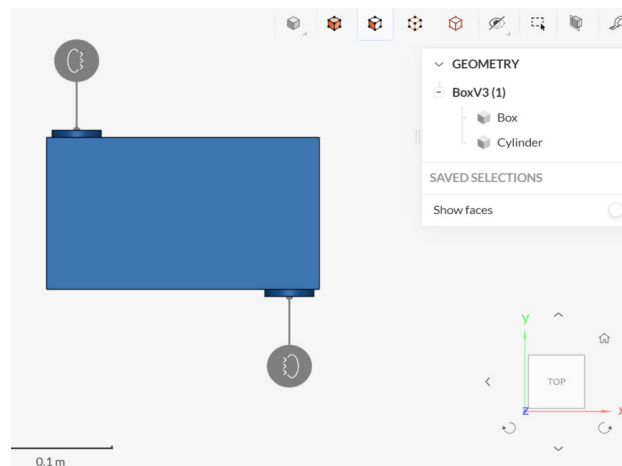
1. To validate the preliminary one-dimensional thermal calculations.
2. To investigate the spatial temperature distribution inside the enclosure.
3. To determine the most suitable location for the temperature sensor.
4. To investigate the airflow generated by the fan.

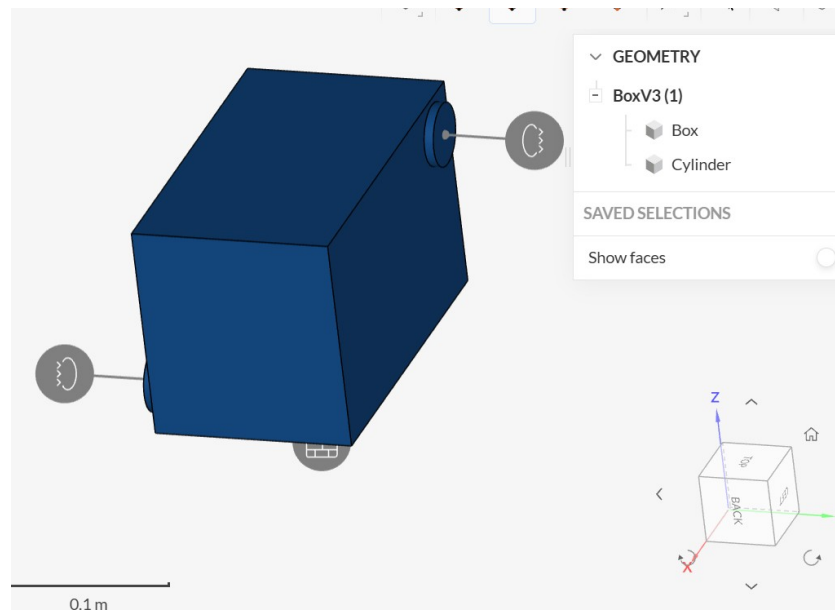
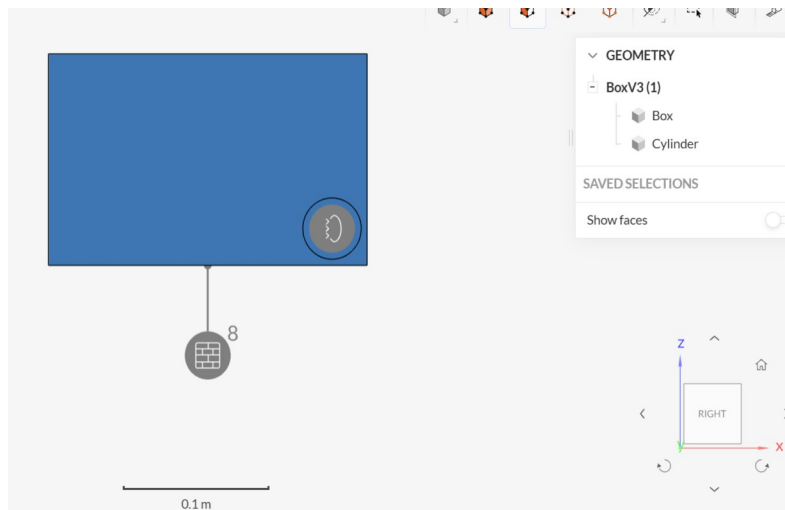
The sensor position must satisfy two competing requirements. First, it should not significantly disturb the airflow. Second, it should measure a temperature representative of the enclosure rather than a highly localized temperature near the heater or airflow outlet.

The enclosure uses two openings located on opposite faces. The inlet is positioned at the bottom of the enclosure, allowing relatively cold ambient air to enter. The fan and outlet are positioned at the top, taking advantage of the natural tendency of hot air to rise and promoting mixing between the forced and buoyancy-driven flows.

The heat source is positioned at the bottom of the enclosure and is modelled in the CFD analysis as a cylindrical region generating a constant heat output.

The CFD analysis indicates that a suitable sensor location is at the centre of the ceiling. At this location the sensor has limited interference with the main airflow and measures a temperature representative of the average enclosure temperature.





Boundary Conditions

The enclosure walls are modelled as cork using the corresponding thermal conductivity:

$$k=0.04 \text{ W/(mK)}$$

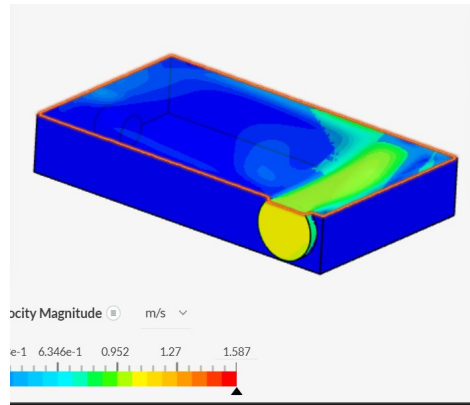
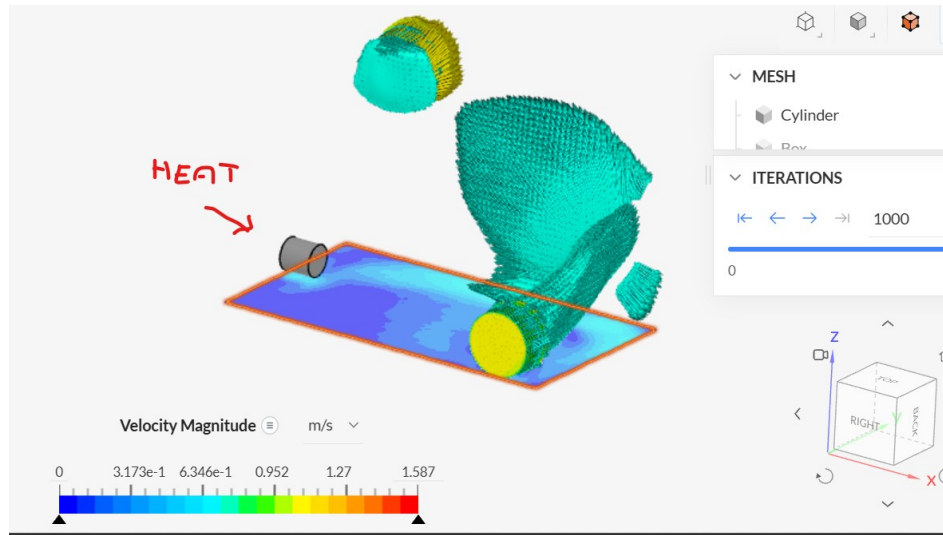
The wall thickness is:

$$L=6 \text{ mm}$$

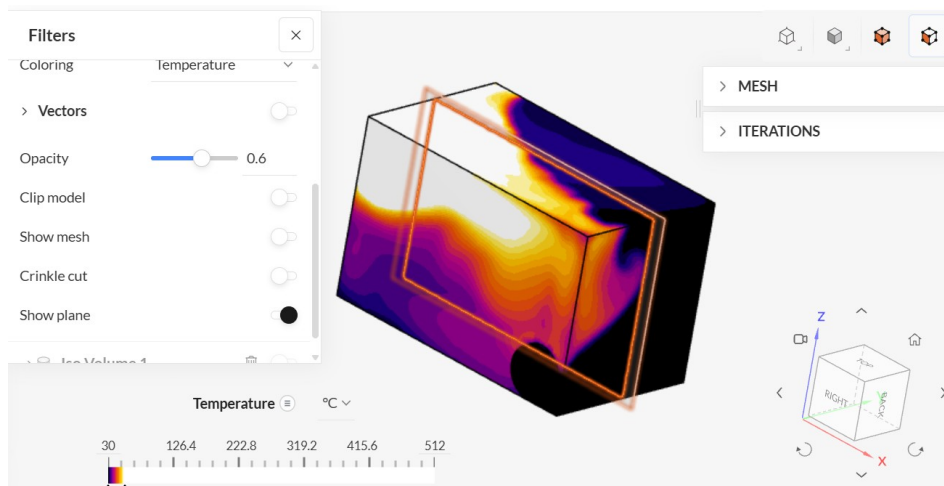
The airflow boundaries are defined as follows:

- Inlet: ambient pressure.
- Outlet with fan: prescribed flow corresponding to the maximum fan airflow.
- Outlet without fan: ambient pressure.

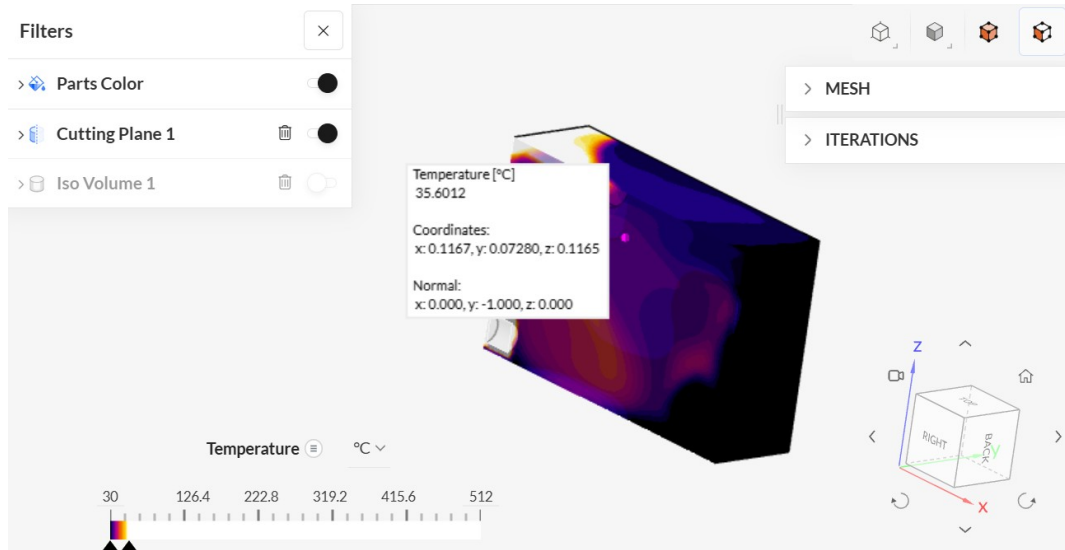
CFD Results — Fan at 100%



The velocity field shows the airflow generated by the fan and the resulting circulation inside the enclosure



The simulated temperature field shows a relatively well-mixed enclosure.

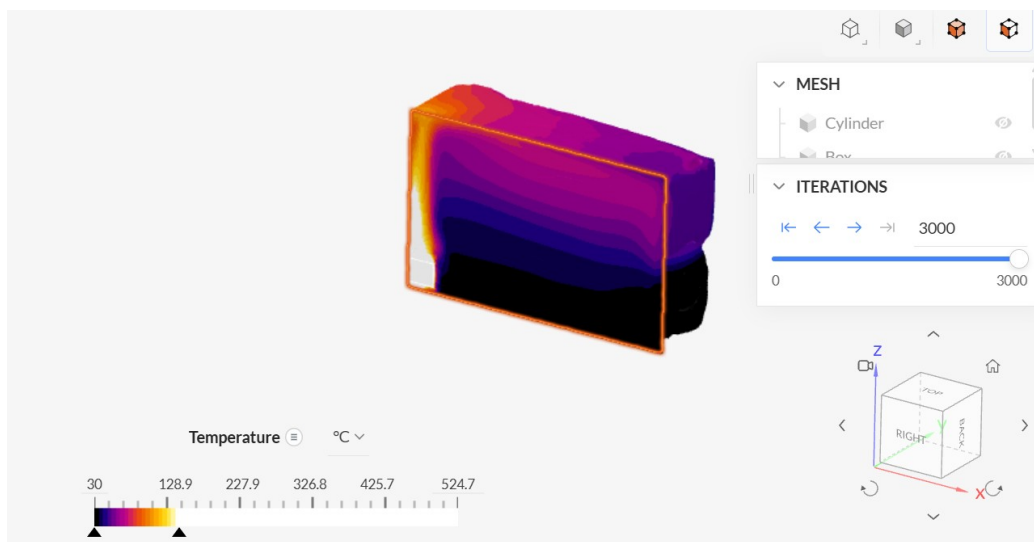


At the selected sensor position, the predicted temperature is about:

$T_{\text{sensor}} = 35.6^{\circ}\text{C}$

This result is reasonably close to the theoretical one-dimensional prediction of 38°C .

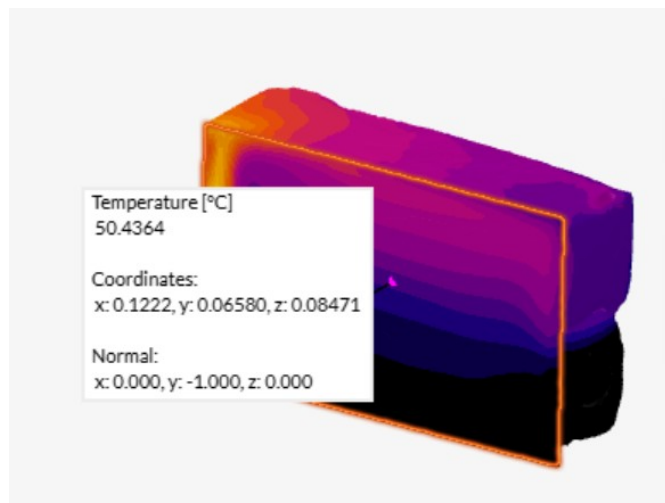
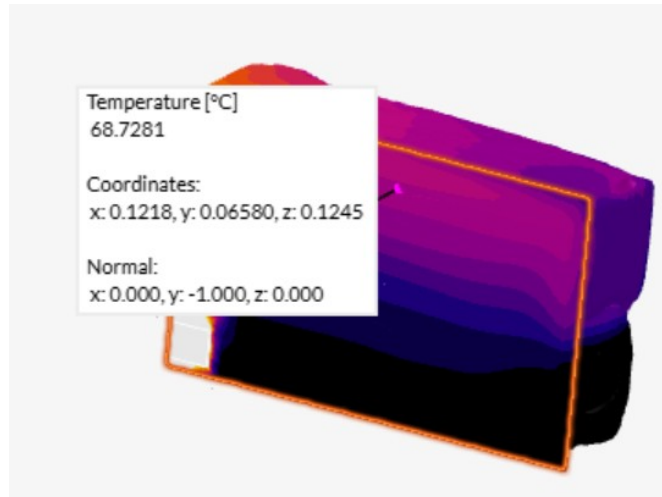
CFD Results — Fan OFF



When the fan is disabled, the CFD model predicts a significantly stronger temperature stratification.

The highest temperatures occur near the heat source, while the temperature around the center of the enclosure is substantially lower (50 C) and close to the 1D energy balance prediction.

The temperature measured at the selected sensor location is around 65 C,

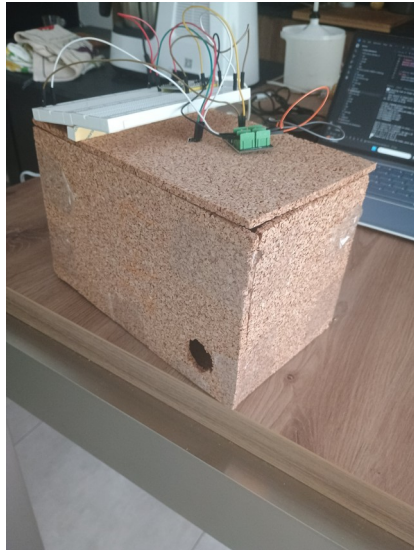


The CFD model predicts local temperatures of around 130°C in the immediate vicinity of the idealized heat source. This result is primarily associated with the modelling of the heat source and its proximity to the enclosure wall.

In the physical implementation, the heat source is not in contact with the electronic components or the sensor. Consequently, this local CFD maximum does not represent the temperature experienced by the components.

3.3 Hardware Implementation

The enclosure is fabricated from cork panels manually cut to the required dimensions.



The temperature sensor is positioned inside the enclosure. Aluminum foil is installed around the sensor to reduce direct radiative heat exchange between the heat source and the sensing element.

The electronics consist of:

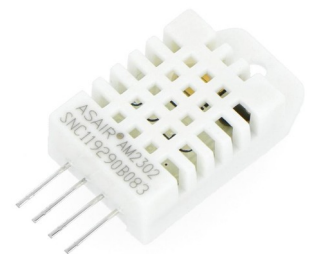
- ESP32 development module
- DHT22 temperature sensor
- 5 V cooling fan
- MOSFET driver
- Breadboard
- Dupont cables
- Constant 15 W heat source

The use of modular components and Dupont connections minimizes the need for soldering and satisfies the plug-and-play objective of REQ-005.

3.4 Temperature Sensor

The selected temperature sensor is a DHT22 / AM2302 digital temperature and humidity sensor.

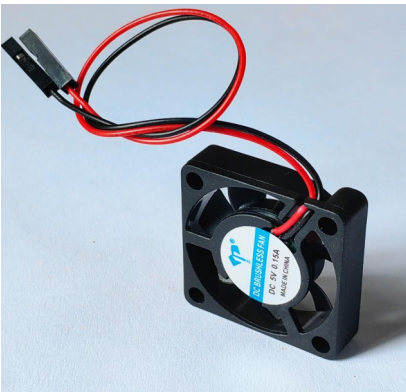
Parameter	Value
Sensor type	Digital temperature and humidity sensor
Temperature range	-40°C to +80°C
Temperature accuracy	±0.5 °C
Temperature resolution	0.1 °C
Humidity range	0–99.9 %RH
Humidity accuracy	±2 %RH
Humidity resolution	0.1 %RH
Supply voltage	3.3–5 V
Interface	Single-wire digital
Minimum measurement interval	2 s
Data output	40-bit digital
Error detection	Checksum



3.5 Cooling Fan

The cooling actuator is a 5 V two-wire fan.

Parameter	Value
Product type	Cooling fan
Connector	2-pin
Supply voltage	5 V
Maximum speed	6500 RPM
Maximum airflow	5 m³/h



Parameter	Value
Noise level	25 dB
Cooling method	Air

The fan can be powered from the available 5 V ESP Pin, simplifying the physical assembly.

3.6 MOSFET

The fan is controlled using a MOSFET connected to a PWM output of the ESP32.

Parameter	Value
Current rating	22 A
Connector type	Screw
Number of contacts	8
Mounting	PCB mount
Upper rated temperature	85 °C



The MOSFET allows the low-power ESP32 GPIO signal to control the higher-current fan supply

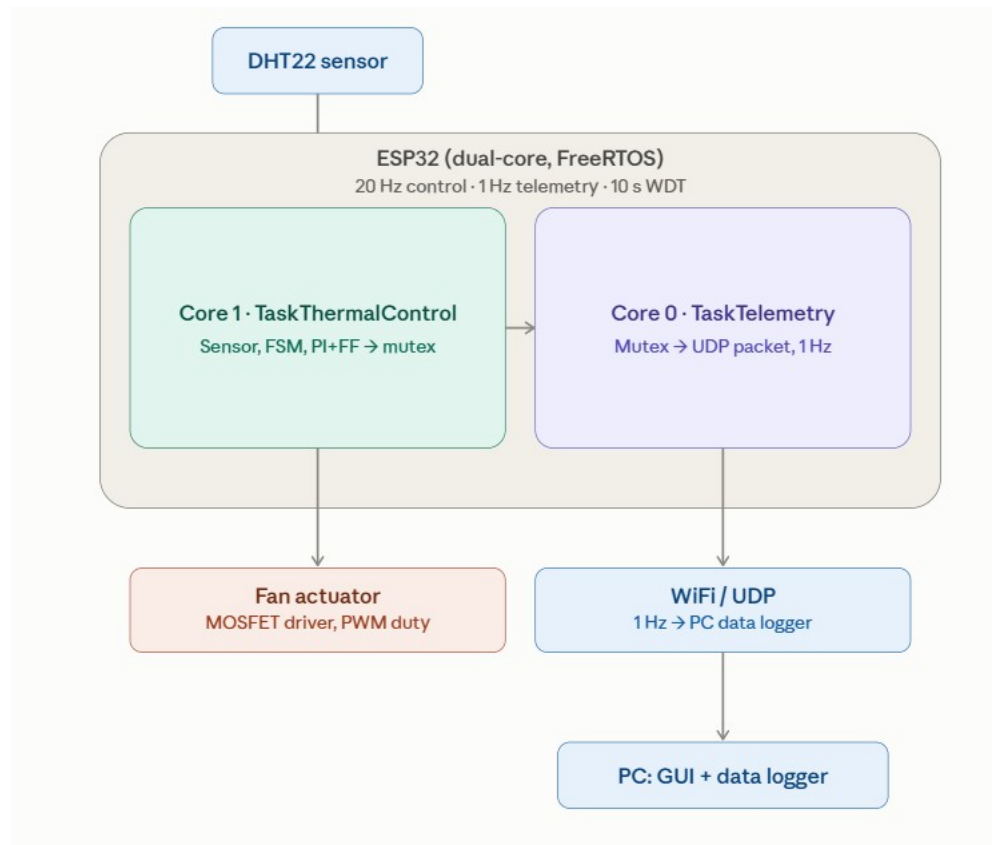
3.7 Software Architecture

The software running on the ESP32 microcontroller implements the active PID thermal regulation loop and the data logging.

This code is implemented as a concurrent, dual-core thermal regulation system using FreeRTOS on the ESP32. It employs a Finite State Machine (FSM) for thermal profile execution, a PI Controller with Feed-Forward (dSP/dt), thread-safe data exchange via Mutexes, and active system health monitoring via the Hardware Watchdog (WDT).

The principal control architecture consists of:

- Finite State Machine (FSM) for thermal profile execution.
- Discrete PI controller.
- Rate-of-change setpoint Feed-Forward.
- Mutex-protected shared telemetry.
- Hardware Watchdog Timer.
- Independent telemetry task.



Task Architecture

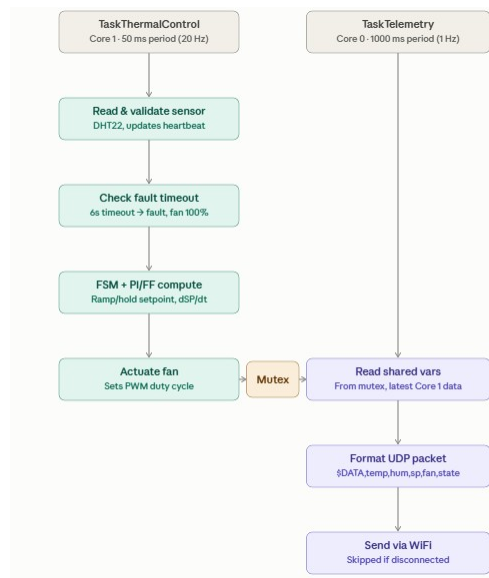
Task	Core	Priority	Period	Responsibility
TaskThermalControl	1	3	50 ms / 20 Hz	Sensor acquisition, FSM update, PI calculation and PWM output
TaskTelemetry	0	2	1000 ms / 1 Hz	Telemetry acquisition, UDP packet generation and Wi-Fi transmission

Both tasks use `vTaskDelayUntil()` rather than `vTaskDelay()`.

This approach provides deterministic periodic execution because the next wake-up time is calculated relative to the task's periodic schedule, reducing timing drift caused by variations in execution time.

The TaskThermalControl task executes at 20 Hz whereas the DHT22 sensor provides valid temperature updates at 0,5 Hz. This multi-rate architecture is intentionally selected for these reasons:

- **Future Scalability:** It preserves architectural flexibility for drop-in upgrades to higher-frequency temperature sensors without altering the core task scheduling or control pipeline.
- **Smooth Setpoint Ramping:** Executing the internal Finite State Machine (FSM) at 20 Hz ensures smooth, continuous evaluation of the setpoint derivative (dS_P/dt) and feed-forward terms during dynamic temperature ramps.
- **Data Gating:** The temperature value fed into the controller (`lastValidTemp`) is refreshed only when a validated sensor frame is received (~ 0.5 Hz), preventing an invalid or stale reading from ever entering the control loop. The PI/Feed-Forward computation itself still executes every cycle (20 Hz), which is required — not just permitted — because the setpoint continues to advance during a ramp and the controller must track it smoothly, as described above.



3.8 Thermal Profile Finite State Machine

The thermal profile is implemented using a finite state machine.

The principal states are:

- STATE_RAMP
- STATE_HOLD
- STATE_FINISHED

During STATE_RAMP, the setpoint is updated according to the specified temperature rate of change:

$dSPdt$

This derivative is also provided to the Feed-Forward component of the controller.

During STATE_HOLD, the setpoint remains constant and:

$dSPdt=0$

After the specified holding time has elapsed, the FSM advances to the next profile step.

Once all profile steps have been completed, the system enters STATE_FINISHED.

The maximum heating and cooling rates are determined experimentally:

$MAX_RAMP_RATE_UP=0.0275^{\circ}C/s$

$MAX_RAMP_RATE_DOWN=0.0160^{\circ}C/s$

The profile rates used during normal operation are deliberately lower than these experimentally measured limits in order to provide sufficient control margin.

An example profile is:

```
const ThermalStep PROFILE[] = {  
    { STEP_RAMP, 38.0f, MAX_RAMP_RATE_UP / 1.5f, 0 },  
    { STEP_HOLD, 38.0f, 0.0f, 3 * 60 * 1000 },/  
    { STEP_RAMP, 42.0f, MAX_RAMP_RATE_UP / 1.5f, 0 }, /  
    { STEP_HOLD, 42.0f, 0.0f, 3 * 60 * 1000 },/  
    { STEP_RAMP, 39.0f, MAX_RAMP_RATE_DOWN / 2.0f, 0 },  
    { STEP_HOLD, 39.0f, 0.0f, 3 * 60 * 1000 },  
    { STEP_RAMP, 37.0f, MAX_RAMP_RATE_DOWN / 2.0f, 0 },  
    { STEP_HOLD, 37.0f, 0.0f, 5 * 60 * 1000 }  
};
```

3.9 PI Controller with Feed-Forward

After a valid DHT22 measurement is obtained, the measured temperature, setpoint, and setpoint derivative are passed to the controller:

$$u[k] = K_p e[k] + K_i \sum_{j=0}^k e[j] \Delta t + K_{FF} \left(\frac{SP[k] - SP[k-1]}{\Delta t} \right)$$

where the PI contribution is based on the temperature error and the Feed-Forward contribution depends on the desired setpoint rate of change.

A 20 Hz task period is used for smooth PWM generation and profile computation. The temperature value used in the error calculation is refreshed only when a validated sensor frame is received, while the PI/Feed-Forward computation itself runs every cycle to track the ramping setpoint smoothly (see 3.7).

The Feed-Forward term is asymmetric because the actuator has only cooling capabilities, infact during a positive setpoint ramp, the controller must compensate for the thermal inertia of the chamber and therefore reduce the effective cooling command. During a negative setpoint ramp, the controller increases the fan command according to the experimentally characterized cooling rate. Because the actuator can only remove heat, the feed-forward term is inherently asymmetric.

During heating, the controller uses the experimentally determined:

MAX_RAMP_RATE_UP

while during cooling it uses:

MAX_RAMP_RATE_DOWN

The integral term is clamped to prevent integral wind-up.

The derivative term is equal to zero as in this specific model the sensor update rate is very slow and the derivative term could work on the noise.

The resulting control command is converted into a fan PWM percentage. Because the fan does not necessarily start reliably at very low PWM values, the controller output is scaled between a minimum effective fan duty cycle and 100%.

3.10 Thread Safety

The two FreeRTOS tasks share several telemetry variables, including:

- measured temperature
- humidity

- setpoint
- fan command
- controller status

These variables are protected using the mutex `xMutexRAM`.

`TaskThermalControl` acts as the principal writer. It acquires the mutex for a short period after acquiring a sensor measurement and updates the shared variables.

`TaskTelemetry` acquires the mutex and immediately copies the shared data into local variables. The mutex is then released before UDP packet formatting and Wi-Fi communication are performed.

This minimizes mutex holding time and avoids unnecessarily blocking the higher-priority thermal-control task.

3.11 Fault-Tolerance Mechanisms

Several mechanisms are implemented to improve system robustness.

Hardware Watchdog

A Hardware Watchdog Timer monitors the execution of the software tasks.

Both tasks register with the watchdog and periodically reset it during normal execution.

If a task becomes permanently blocked or the software stops executing correctly, the watchdog eventually triggers a system reset.

The implemented watchdog timeout is:

TWDT=10 s

Sensor Fault Handling

During normal operation, invalid DHT22 measurements are detected through the sensor validation mechanism.

If a temperature reading is invalid or unavailable, the control software transitions to the fault state and commands the fan to its safe-state output of 100%.

The initialization phase allows several attempts to establish communication with the sensor before declaring a fault.

For safety purposes, any invalid sensor value should not be interpreted as a valid temperature measurement.

Network Isolation

The telemetry task is independent from the thermal-control task.

Before attempting UDP transmission, the software checks whether Wi-Fi is connected.

If communication is unavailable, the telemetry task skips transmission while the thermal-control task continues to operate.

Therefore, loss of network connectivity cannot directly stop the thermal-control loop.

4. Verification and Validation

The experimental campaign has three main objectives:

1. Determine the maximum steady-state temperature and temperature rate of change for fan-off and fan-on conditions.
2. Validate the experimentally selected heating and cooling ramp rates.
3. Validate the ability of the controller to follow a dynamic temperature profile.

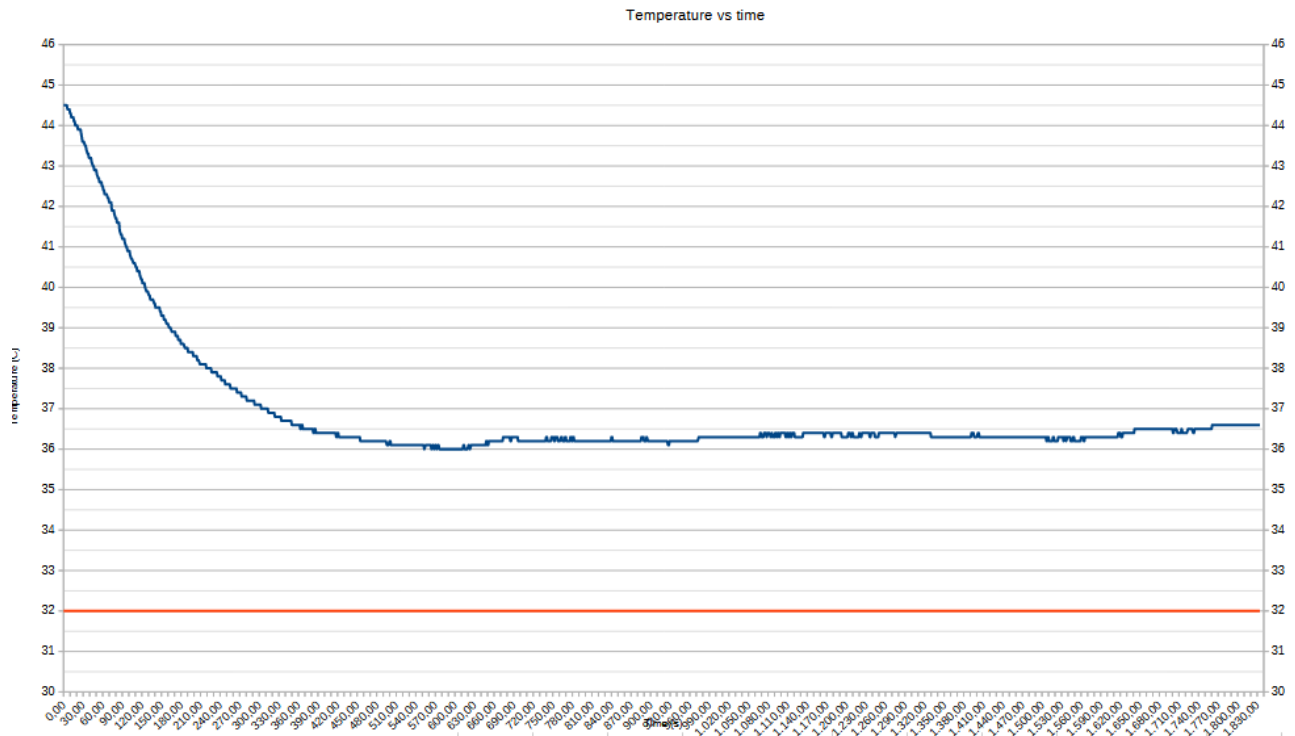
The tests also provide evidence for the thermal assumptions used in the system safety argument.

4.1 TEST-01 — Fan at 100%

The first test evaluates the thermal behaviour with the fan operating at 100% PWM.

The setpoint is maintained at 32°C, while the system is allowed to reach its maximum operating temperature.

The ambient temperature is $T_{amb}=27.5^{\circ}\text{C}$



The measured results are:

$T_{steadystate}=36\pm0.5^{\circ}\text{C}$

The one-dimensional theoretical model predicts :

Tsteadystate theory=35.5°C

The CFD simulation, extrapolated from the simulation performed at an ambient temperature of 30°C, gives approximately 32.5°C.

The experimental result therefore agrees reasonably well with the analytical model.

The experimentally determined cooling rate is calculated over the relevant temperature interval:

$$dT/dt = (44.5-36)/540 \approx 0.0157 \text{ } ^\circ\text{C/s}$$

which is approximately:

$$0.016 \text{ } ^\circ\text{C/s}$$

Therefore:

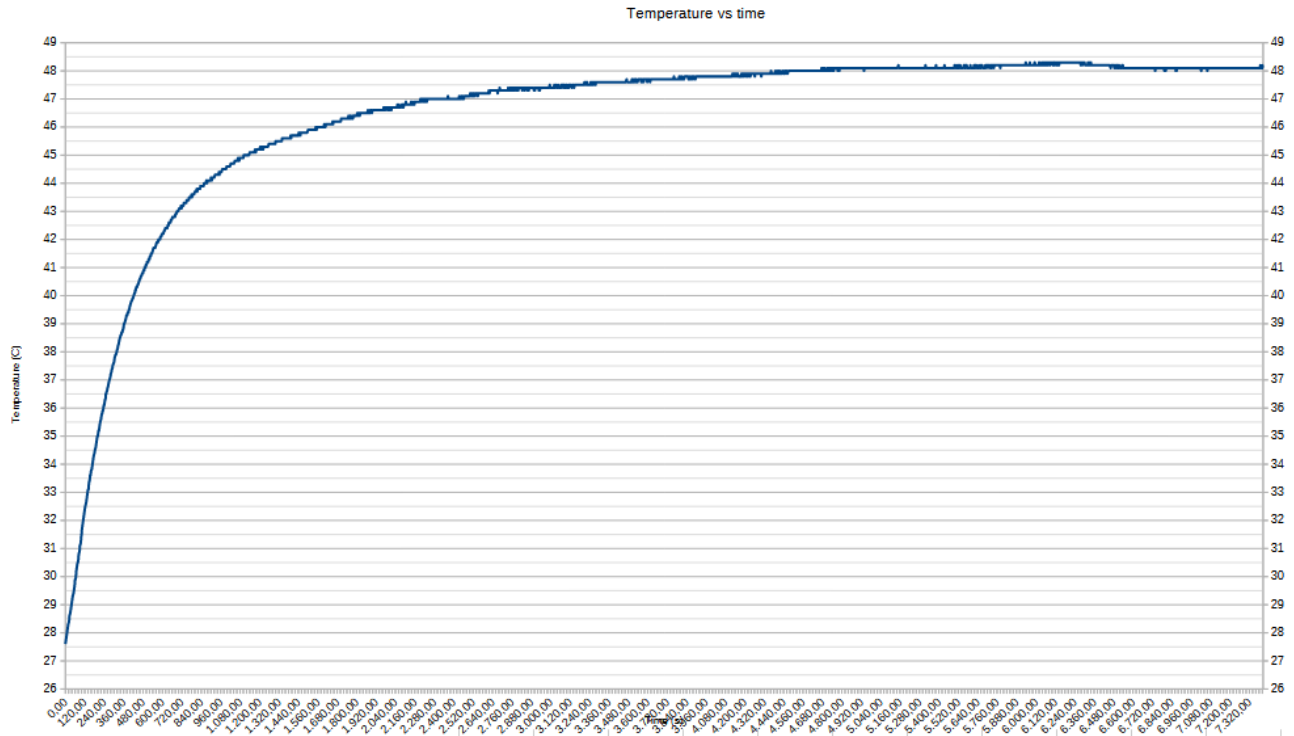
$$\text{MAX_RAMP_RATE_DOWN} = 0.016 \text{ } ^\circ\text{C/s}$$

This value is used as the maximum experimentally demonstrated cooling rate in the controller.

4.2 TEST-02 — Fan OFF

The second test evaluates the thermal behaviour with the fan disabled.

The heat source is allowed to increase the enclosure temperature from an ambient temperature of $T_{amb}=27.5^{\circ}\text{C}$



The measured steady-state temperature is:

$$T_{\text{steadystate}} = 48 \pm 0.5^{\circ}\text{C}$$

The one-dimensional analytical model predicts:

$$T_{\text{steadystate theory}} = 46.5^{\circ}\text{C}$$

The CFD model, extrapolated from the $T_{amb} = 30^{\circ}\text{C}$ simulation, predicts a substantially higher value of approximately 62°C at the sensor location.

The lower experimental temperature can be explained by differences between the idealized CFD model and the physical enclosure. In particular, the CFD model the enclosure is modelled as perfectly sealed/insulated, in the real test this is not true and additional air exchange with the external environment provides an additional heat-transfer mechanism giving a less important temperature stratification.

The maximum heating rate is calculated over a 10°C temperature interval:

$$dT/dt = (37.5 - 27.5)/360 = 0.0278^{\circ}\text{C/s}$$

This is close to the implemented value:

MAX_RAMP_RATE_UP = 0.0275°C/s

A 10°C interval is used because this is representative of the temperature variations encountered by the control loop; using a substantially larger interval would not necessarily provide a meaningful representation of the instantaneous thermal ramp rate.

4.3 TEST-03 — Dynamic Setpoint Tracking

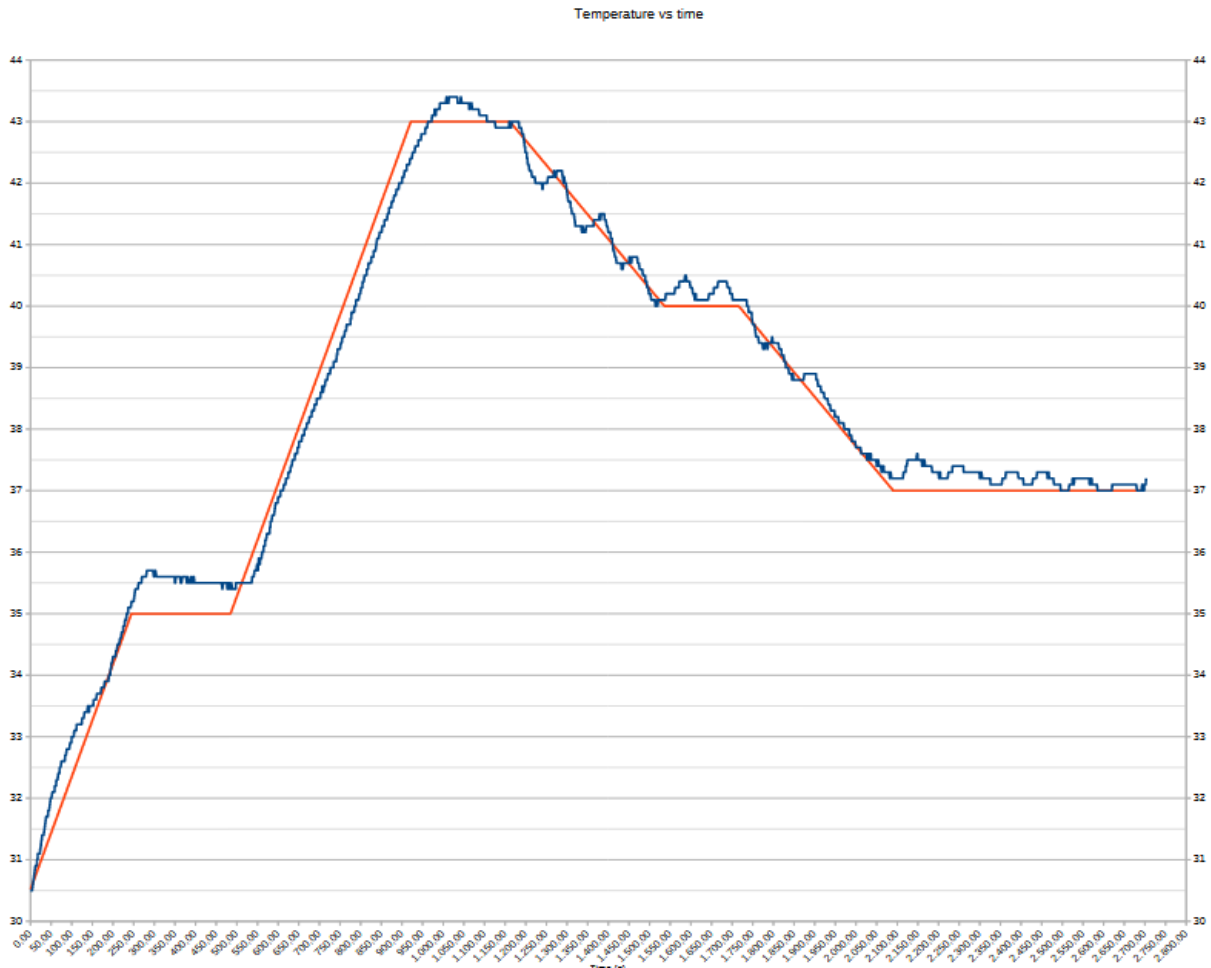
The third test evaluates the ability of the control system to follow a dynamic temperature profile.

The profile consists of:

1. Heat to 35°C
2. Hold at 35°C for 4 minutes.
3. Heat to 43°C.
4. Hold at 43°C for 4 minutes.
5. Cool to 40°C.
6. Hold at 40°C for 3 minutes.
7. Cool to 37°C.
8. Hold at 37°C for 5 minutes.

The ramp rates are set to fractions of the experimentally determined maximum heating and cooling rates.

The test is performed with a starting temperature < 35°C



The maximum absolute tracking error is between 0.5°C and 0.7°C.

Around 50 measurements exceed the sensor accuracy threshold out of a total of 2760 measurements.

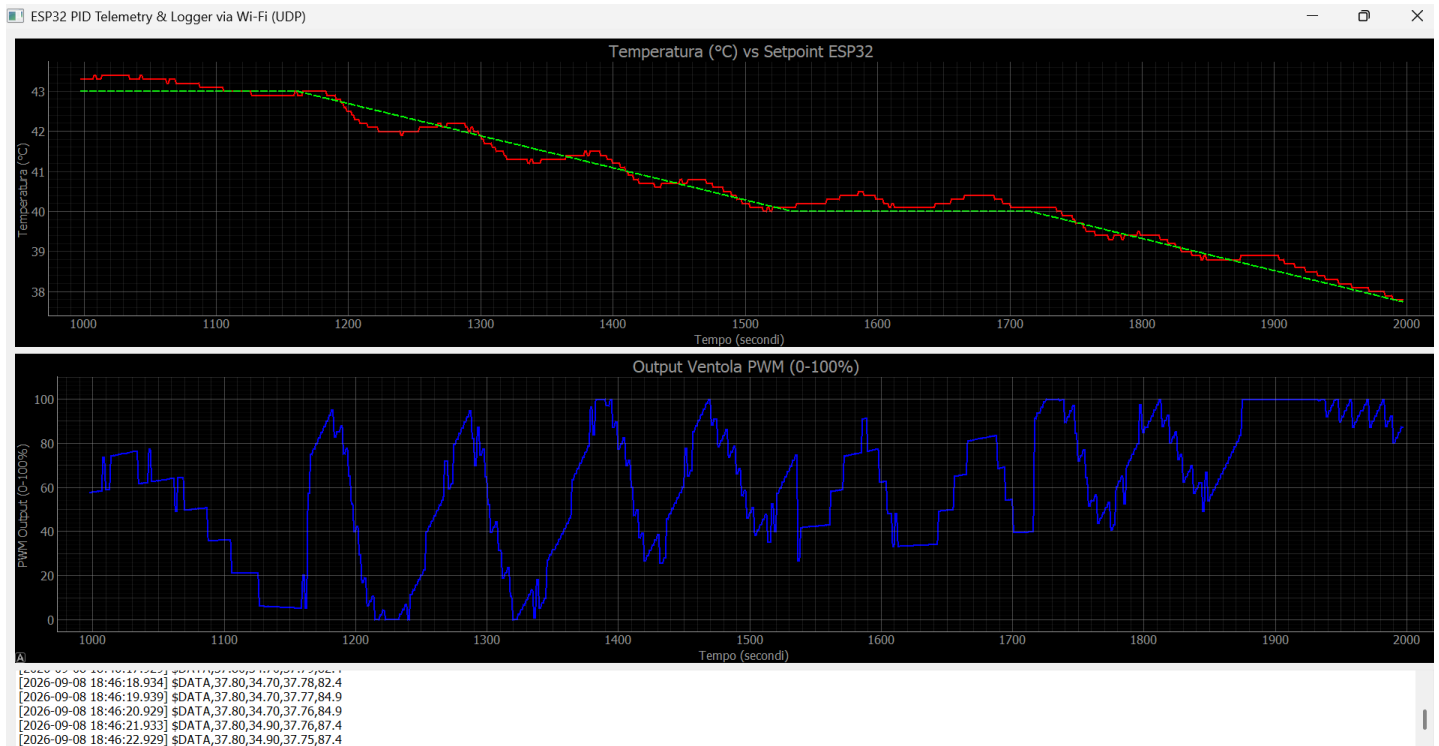
Therefore:

$$50/2760 \times 100 \approx 1.81\%$$

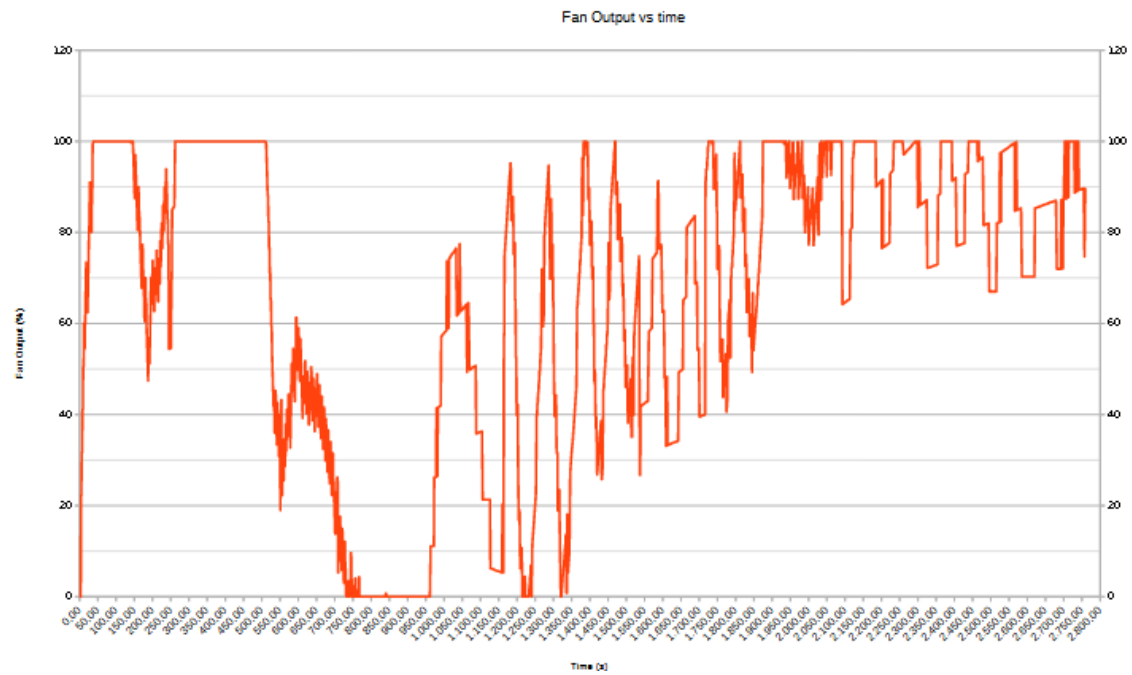
Thus, **1.8% of the measured samples exceed the nominal sensor accuracy by up to 0.2°C.**

The results demonstrate that the controller is capable of tracking the dynamic profile with a relatively small tracking error.

The temperature, setpoint, and fan PWM are transmitted in real time through the telemetry task and displayed by a Python-based graphical interface.



The interface also provides a data log that can be exported for subsequent post-processing.



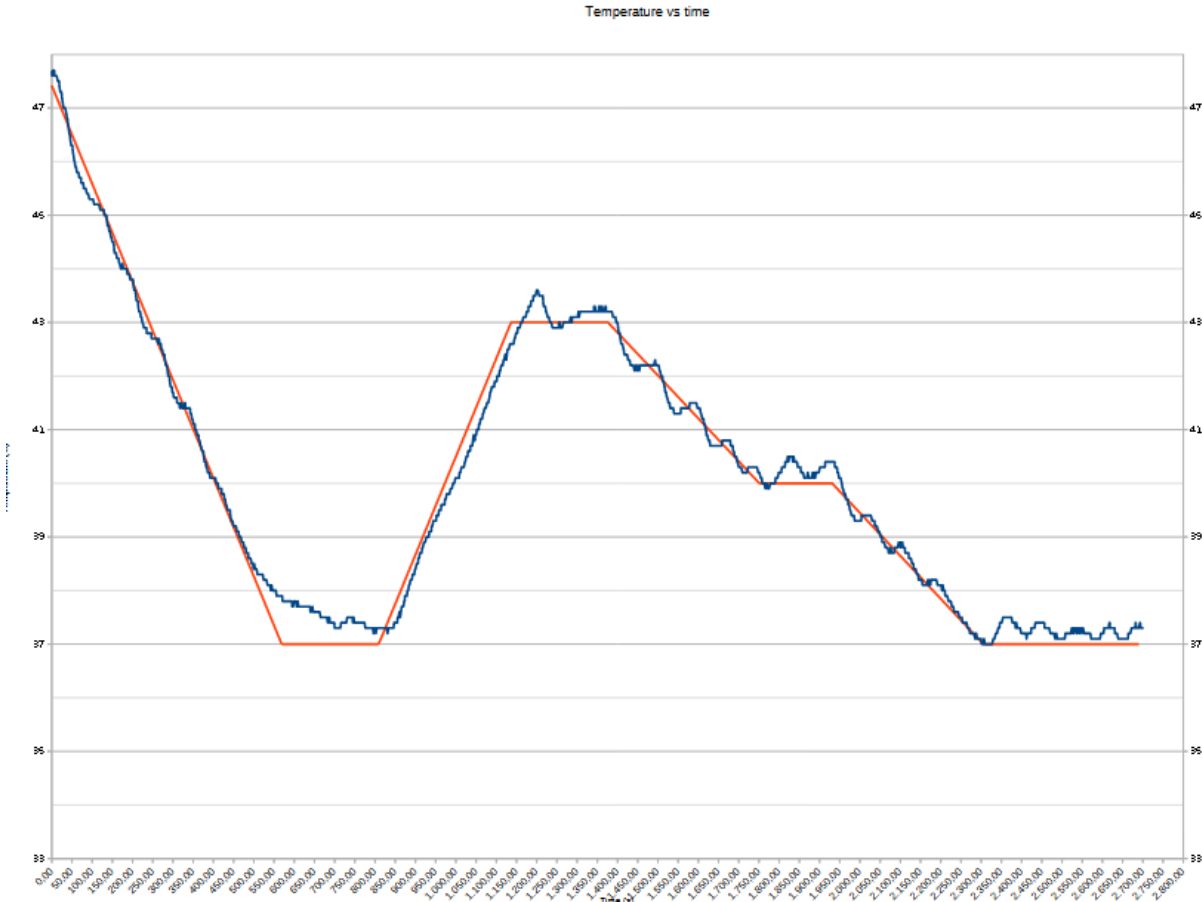
The average fan output during the test is approximately 67%

This indicates that the fan does not remain saturated at 100% and that sufficient actuator margin remains available during normal profile tracking.

4.4 TEST-04 — Dynamic Cooling from Elevated Initial Temperature

The fourth test uses the same dynamic profile but starts from an elevated initial temperature of approximately 47°C

The profile therefore begins with a cooling phase toward 35°C, using the same maximum cooling rate defined during TEST-01.



The maximum absolute tracking error is between :

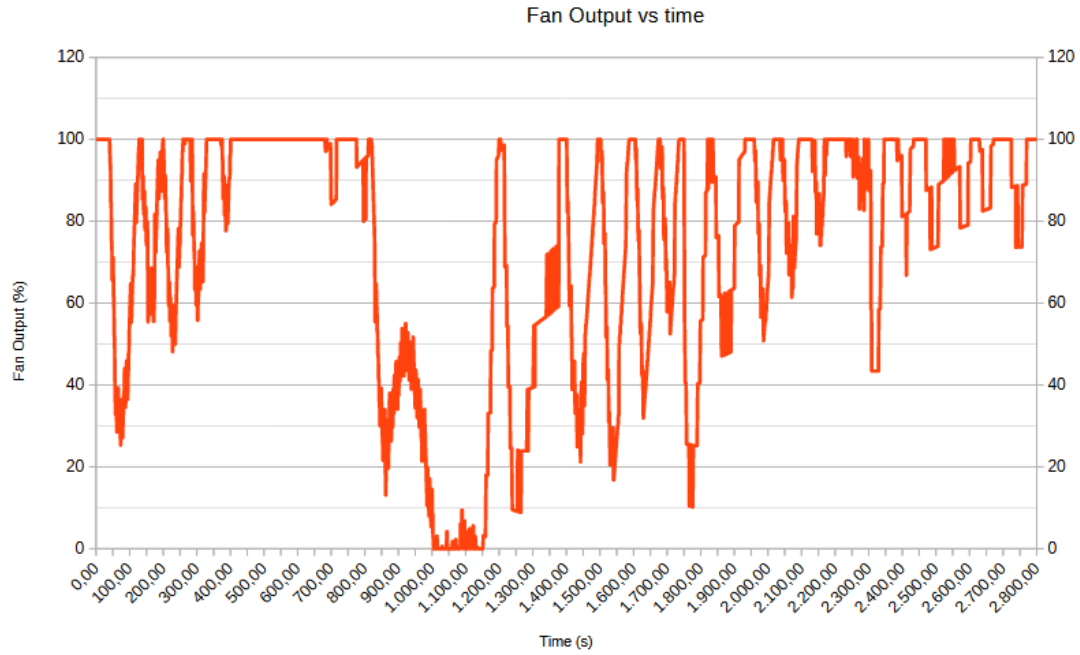
0.5°C and 0.9°C

Approximately 150 measurements exceed the nominal sensor accuracy out of 2760 measurements.

The corresponding percentage is:

$$150/2760 \times 100 \approx 5.43\%$$

Therefore, **5.4% of the measurements exceed the nominal sensor accuracy**, with an maximum error of up to 0.4°C.



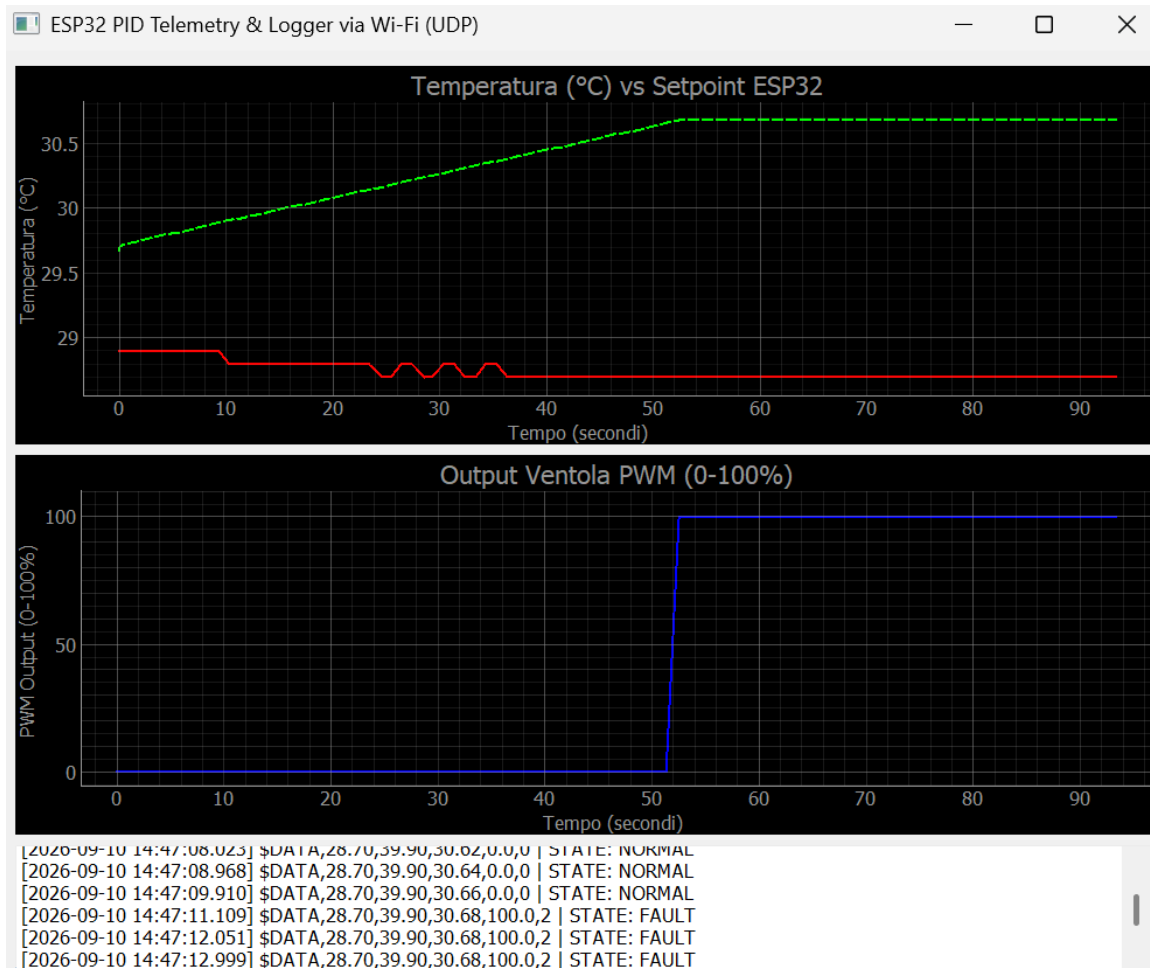
The average fan output is around 73%.

The fan therefore retains some control margin and does not remain continuously saturated during the test.

The larger tracking error compared with TEST-03 is consistent with the more demanding initial thermal condition.

4.5 TEST-05 — Fault injection and safe state transitions

The objective of this test is to verify the implementation of SR-01, SR-02, SR-04, and SR-06 by simulating a physical sensor failure during active closed-loop execution.



The GUI we can see the transition to the STATE: Fault and the Fan output jumping to 100%. In the log on the serial monitor we can see that after a data age of 6050 ms of incorrect sensor data there is the switch to a safe state.

```
HEARTBEAT] Now: 80315 ms | LastValidTS: 79015 ms | Data Age: 1300 ms / 6000 ms limit
[SENSOR DEBUG] Valid read @ 81065 ms | Temp: 28.70 °C | LastValidTS: 81065 ms
[HEARTBEAT] Now: 81315 ms | LastValidTS: 81065 ms | Data Age: 250 ms / 6000 ms limit
[HEARTBEAT] Now: 82315 ms | LastValidTS: 81065 ms | Data Age: 1250 ms / 6000 ms limit
[HEARTBEAT] Now: 83315 ms | LastValidTS: 81065 ms | Data Age: 2250 ms / 6000 ms limit
[HEARTBEAT] Now: 84315 ms | LastValidTS: 81065 ms | Data Age: 3250 ms / 6000 ms limit
[HEARTBEAT] Now: 85315 ms | LastValidTS: 81065 ms | Data Age: 4250 ms / 6000 ms limit
[HEARTBEAT] Now: 86315 ms | LastValidTS: 81065 ms | Data Age: 5250 ms / 6000 ms limit
[SAFETY CRITICAL] Sensor timeout reached! Data age (6050 ms) > Limit (6000 ms). Entering STATE_FAULT!
[HEARTBEAT] Now: 87315 ms | LastValidTS: 81065 ms | Data Age: 6250 ms / 6000 ms limit
[HEARTBEAT] Now: 88315 ms | LastValidTS: 81065 ms | Data Age: 7250 ms / 6000 ms limit
[HEARTBEAT] Now: 89315 ms | LastValidTS: 81065 ms | Data Age: 8250 ms / 6000 ms limit
[HEARTBEAT] Now: 90315 ms | LastValidTS: 81065 ms | Data Age: 9250 ms / 6000 ms limit
[HEARTBEAT] Now: 91315 ms | LastValidTS: 81065 ms | Data Age: 10250 ms / 6000 ms limit
```

4.6 Requirement-to-Test Traceability

The final verification matrix links system requirements and safety requirements to the test

Requirement ID	Requirement Description	Type	Design Element	Implementation Element	Verification Method	Pass / Fail
REQ-001	Maintain enclosure temperature within an acceptable range of a constant setpoint.	Functional	Discrete PI Controller with Feed-Forward	TaskThermalControl, PWM Fan Control	TEST-01, TEST-03 (Hold phases)	PASS – Tracking error < 0.5 C
REQ-002	Track dynamic temperature setpoints within an acceptable error range.	Functional	Finite State Machine (FSM) & Asymmetric Feed-Forward	dSPdt calculation, STATE_RAMP	TEST-03, TEST-04	PASS – Tracking error within 1 C
REQ-003	Provide real-time graphical representation of temperature, setpoint, and actuator utilization.	Functional	Independent Telemetry Task & Wi-Fi Link	TaskTelemetry, UDP transmission, Python GUI	TEST-03, TEST-04	PASS
REQ-004	Provide measured temperature and actuator utilization in tabular form for logging/analyses.	Functional	PC Data Logger	Data logging interface / CSV export	TEST-03, TEST-04	PASS
REQ-005	Modular assembly using plug-and-play components to minimize permanent connections/soldering.	Assembly	Commercial off-the-shelf (COTS) modular hardware	Breadboard, Dupont cabling, Screw-terminal MOSFET	Inspection / Physical Assembly	PASS
SR-01	Periodic temperature monitoring with intervals not exceeding sensor limits.	Safety	DHT22 periodic acquisition	TaskThermalControl (20 Hz execution, 2s sensor rate)	TEST-05 (Fault Injection)	PASS – Timeout triggered at 6050 ms

Requirement ID	Requirement Description	Type	Design Element	Implementation Element	Verification Method	Pass / Fail
SR-02	Detect loss or invalidity of temperature measurements and transition to STATE_FAULT.	Safety	Data validation & Checksum verification	Sensor fault-handling routine	TEST-05 (Fault Injection)	PASS
SR-03	Cooling actuator failure shall not cause violation of Tmax	Safety	Passive thermal boundary design	Cork enclosure, natural heat dissipation	TEST-02 (Fan-OFF steady-state testing)	PASS – Tmax < 80 C without fan thanks to the enclosure thermal dynamic
SR-04	Command fan to 100% duty cycle safe-state on critical faults	Safety	Software Fault Handler & Hardware Watchdog	ESP32 WDT = 10 s, Safe-state override	TEST-05 (Fault Injection)	PASS — FSM STATE_FAULT and 100% Fan command
SR-05	Independent passive thermal protection preventing temperature from exceeding Tmax	Safety	Passive heat transfer envelope	Cork box dimensions and wall thermal resistance	TEST-02	PASS - Enclosure dissipates heat independently of the control system at the ambient conditions
SR-06	Fault detection and response time compatibility with enclosure thermal dynamics.	Safety	Deterministic task scheduling	vTaskDelayUntil() implementation on Core 1	TEST-05 (Fault Injection)	PASS — fault detection logic in around 6 s and reaction in 2 s

5. Discussion

The combined analytical, numerical, and experimental results demonstrate that the proposed thermal-control architecture is capable of regulating the enclosure temperature over the intended operating range.

The one-dimensional thermal model provides a simple first-order representation of the enclosure behaviour and gives predictions reasonably close to the experimental results. For

the fan-on case, the measured steady-state temperature of approximately 36°C is close to the analytical prediction of 35.5°C.

The CFD analysis provides additional information that cannot be obtained from the one-dimensional model, particularly regarding airflow, temperature stratification, and the location of local temperature maxima. It also allows the temperature sensor position to be selected based on the internal flow field.

The fan-off CFD case shows a significantly higher temperature than the experiment. This difference highlights the limitations of an idealized CFD model when compared with a manually fabricated enclosure. Real-world leakage, imperfect insulation, and additional air exchange with the environment can significantly modify the thermal behaviour.

The experimental results also demonstrate the importance of distinguishing between **local maximum temperature** and **representative enclosure temperature**. The CFD model predicts high temperatures in the immediate vicinity of the idealized heat source, while the sensor is intentionally located away from this region to provide more average temperature of the enclosure.

From a control perspective, the measured maximum heating and cooling rates provide experimentally derived boundaries for the dynamic setpoint generator. The profile rates are intentionally selected below these boundaries, allowing the PI controller and Feed-Forward mechanism to track the requested profile without requiring continuous actuator saturation.

The two dynamic-profile tests demonstrate that the system can track both heating and cooling trajectories. The average fan utilization of 67% in TEST-03 and 73% in TEST-04 confirms that the actuator retains some available control authority during normal operation.

6. Safety Assessment

The safety analysis combines the top-down hazard analysis with the bottom-up FMEA.

The principal hazard is uncontrolled overheating. The corresponding Safety Goal, SG-01, is to prevent the protected enclosure/component temperature from exceeding TMAX.

The analysis identifies three principal active-system failure modes:

- Temperature sensor failure.
- Fan failure.
- Control-unit/software failure.

Each failure can degrade the normal control function. However, within the defined indoor operating envelope, the passive thermal behaviour of the enclosure provides an independent limit on the resulting temperature.

The active mechanisms — sensor validation, fault-state operation, and watchdog reset — therefore provide additional robustness and improve the availability of the control system, while the passive thermal behaviour provides the fundamental independent protection against the identified overheating hazard.

The safety argument is valid only within the defined environmental boundary conditions. In particular, the passive safety margin decreases as the ambient temperature increases. Operation outside the specified indoor environmental envelope is therefore excluded from the safety claim.

7. Conclusion

This project presented the design, implementation, and experimental validation of a small actively controlled thermal enclosure based on an ESP32 microcontroller.

The system combines a discrete PI temperature controller with rate-of-change Feed-Forward, a finite-state-machine-based thermal profile generator, real-time telemetry, and fault-tolerance mechanisms.

The development process follows a Systems Engineering approach in which requirements, safety objectives, design elements, software elements, and verification activities are explicitly traced.

A tailored functional-safety methodology inspired by IEC 61508 principles was applied. The hazard analysis identified uncontrolled overheating as the principal hazard, from which one Safety Goal and six Safety Requirements were derived. A subsequent FMEA showed that the main identified component failures do not compromise the safety goal within the defined environmental operating envelope because of the independent passive thermal behaviour of the enclosure.

The thermal design was investigated using both one-dimensional analytical models and three-dimensional CFD simulations. The CFD analysis additionally provided information about temperature stratification and airflow, allowing the temperature sensor to be positioned in a representative region of the enclosure.

Experimental tests confirmed the main thermal characteristics of the system. With the fan operating at 100%, the measured steady-state temperature was approximately 36°C, close to the analytical prediction of 35.5°C. With the fan disabled, the measured steady-state temperature was 48°C compared with 46.5°C predicted by the analytical model.

The experimentally determined maximum heating and cooling rates were approximately:

$\text{MAX_RAMP_RATE_UP}=0.0275^{\circ}\text{C/s}$

and

$\text{MAX_RAMP_RATE_DOWN}=0.0160^{\circ}\text{C/s}$

These values were subsequently used to constrain the dynamic setpoint profiles.

The dynamic-profile experiments demonstrated successful tracking of both heating and cooling trajectories. Under the nominal initial-temperature condition, 1.8% of the measurements exceeded the nominal sensor accuracy by a maximum additional error of 0.2°C. Under the more demanding elevated-temperature initial condition, 5.4% of measurements exceeded the nominal sensor accuracy, with an additional error of up to 0.4°C.

Overall, the results demonstrate that the proposed architecture satisfies the principal functional requirements and provides a robust basis for thermal regulation within the defined operating envelope. The combination of active control, experimental characterization, analytical modelling, CFD analysis, and passive thermal protection provides a coherent Systems Engineering approach for a small-scale thermal-control prototype.